



New process to remove (legacy) additives from dissolved polymeric waste with liquid-liquid extraction

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ABSTRACT

Hexabromocyclododecane (HBCD) is a legacy brominated flame retardant historically used in polystyrene insulation materials. Although its production and use have been banned since 2015 in the EU, HBCD-containing products remain in circulation, which requires removal during recycling. This study investigates the removal of HBCD from real polystyrene waste via liquid-liquid extraction during dissolution recycling. Following a comprehensive solvent screening, cyclohexene was identified as an effective solvent for dissolving polystyrene, while dimethyl sulfoxide was selected as the extractant solvent. This solvent pair not only enables efficient HBCD removal but also prevents undesired polymer precipitation during the extraction process. The feasibility of the method was demonstrated using a real extruded polystyrene waste sample, achieving high extraction efficiencies exceeding 93%. Extraction column simulations using Aspen Plus further indicate that a minimum of three theoretical stages and a solvent-to-feed ratio of 0.9 are required to reduce HBCD concentrations below the regulatory limit of 100 ppm in recycled PS. These results provide key design parameters for scaling the process and highlight the potential of this approach for producing safer recycled polystyrene.

1. Introduction

Despite global efforts, plastic recycling rates remain low worldwide, with recycled material accounting for less than 10% of the total plastic produced globally in 2024 [1]. The complexity of plastic waste makes full circularity difficult, because achieving cost-effective closed-loop recycling depends heavily on the quality and purity of the waste stream [2]. One of these complexities is the additive package incorporated into the polymer, which is required to tailor the material's properties and ensure it meets specific application and performance requirements [3]. This is the case for flame retardants, which are typically added to the polymer to inhibit or delay the spread of fire [4]. Flame retardants inhibit combustion by chemically disrupting the flame chemistry, frequently through scavenging high-energy gas-phase radicals that drive flame propagation [4]. Alternatively, some systems promote char formation, producing a protective barrier that restricts heat and oxygen transport. Common examples include brominated flame retardants, phosphorus-based compounds, and antimony synergists [4–7].

These additives can hinder recycling because some are hazardous

substances that limit mechanical recycling, while in chemical recycling the hetero-atoms present on these molecules may deactivate or poison catalysts during processes such as catalytic pyrolysis, or even cause corrosion of processing equipment [3,6]. One of the problematic brominated flame retardants is hexabromocyclododecane (HBCD), commonly found in extruded and expanded polystyrene foams used as insulation boards [8]. Since 2015, HBCD has been classified as a substance of very high concern under REACH legislation as it persists in the environment, is bioaccumulative in biological systems and toxic [8–10]. Even though its production and usage has been prohibited, HBCD has been present in the market since the 1960s and can also be found in other applications such as textiles and electronics [6,8]. As a result, HBCD remains present in older products as legacy chemical due to the long service lifetimes of these applications, and its persistence poses a major challenge for recycling because materials containing HBCD no longer meet regulatory requirements, limiting their reuse and highlighting the need for effective solutions to manage HBCD-containing waste [6,8].

Different strategies have been investigated to remove HBCD from

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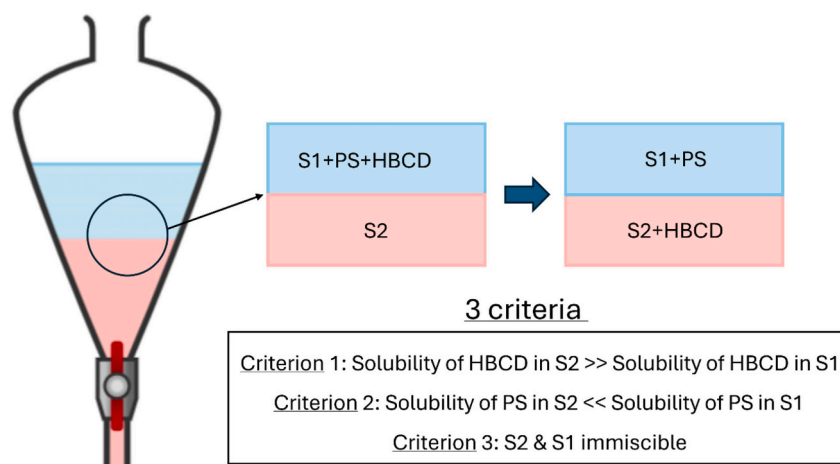


Fig. 1. Schematic overview of liquid-liquid extraction and the three required criteria for successful extraction, with S1 being the solvent and S2 being the extractant solvent.

plastic waste, although most extraction studies focus on analytical methods aimed at quantifying HBCD in various matrices [9,11,12]. For instance, Vilaplana et al. [9] compared the extraction of HBCD from styrene polymeric matrices using pressurized liquid extraction, microwave-assisted extraction and ultrasonic-assisted extraction, and obtained high extraction efficiencies (> 90%) with some of these techniques. These approaches extract HBCD without polymer dissolution and were optimized for analytical purposes rather than for full process design. Soxhlet extraction is also a widely applied technology to remove HBCD from polymeric matrices [12], however, this equipment is not directly scalable as it typically requires high extraction times and energy [3]. Mosallaei et al. [13] investigated the removal of HBCD from electric and electronic equipment with supercritical CO₂/co-solvent mixture, obtaining approximately 55% removal efficiency. The authors highlight that challenges remain in scaling up the process and in achieving sufficient flame retardant removal efficiency to meet regulatory requirements.

A promising approach for removing additives is dissolution recycling, where the polymer is dissolved in a suitable solvent, releasing the

additive package into the solvent and enabling subsequent purification and separation techniques. In our recent study [6], we demonstrated the successful removal of HBCD with adsorption using activated charcoal, with obtained removal efficiencies up to 95%. In this work, the removal of HBCD from polystyrene insulation boards using liquid-liquid extraction is investigated, which to the best of our knowledge has not been applied in dissolution recycling. The goal is to keep polystyrene dissolved in a good solvent, such as cyclohexene as reported in our previous study [14], and extract HBCD with an extractant solvent that will not act as an antisolvent for polystyrene. Combining the required properties, i.e. good affinity to HBCD, immiscibility with cyclohexene, and sufficient affinity for PS not to act as antisolvent, a solvent screening was performed by the Conductor like Screening Model for Realistic Solvents (COSMO-RS) model. After initial solvent selection, the liquid-liquid phase diagram was constructed. Finally, Aspen Plus simulations were performed to identify the number of stages needed to achieve the legal limit of 100 ppm of HBCD in the recycled polymer applicable to its use in the production of expanded and extruded polystyrene insulation material for use in buildings or civil engineering works [15].

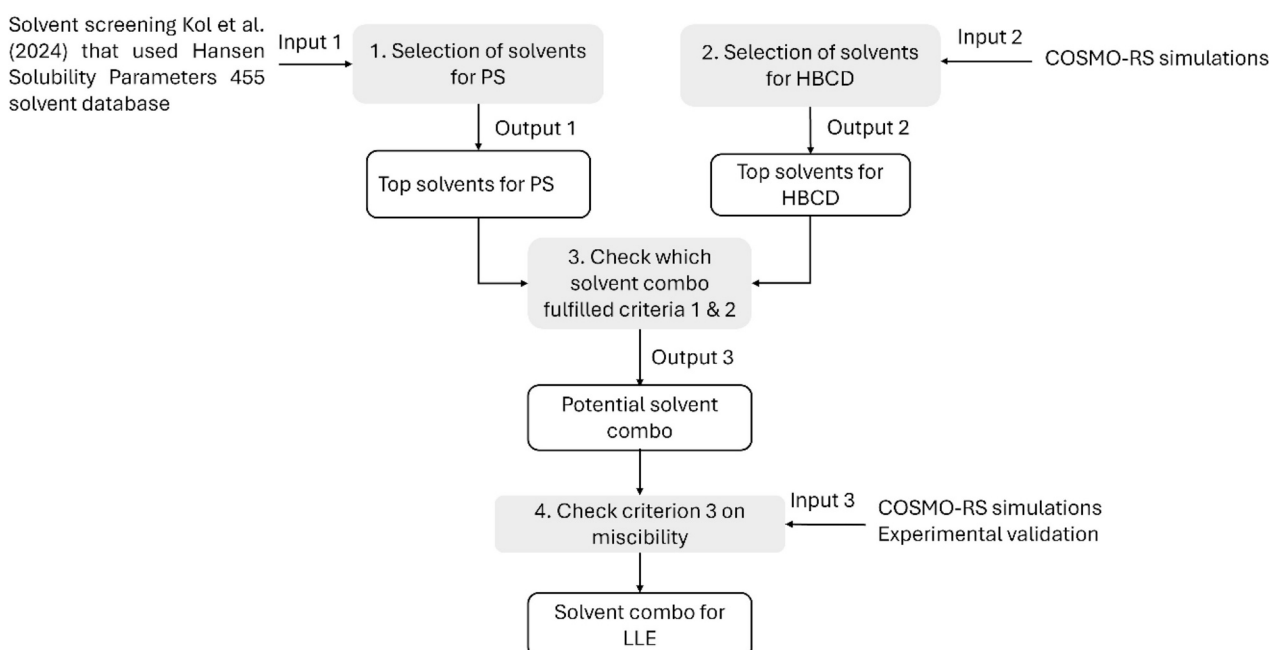


Fig. 2. Methodology for LLE solvent selection.

2. Materials & methods

2.1. Materials

Polystyrene pellets (General Purpose Polystyrene “GPPS”: Styrolution PS 168 N) were provided by INEOS Styrolution. The solvents cyclohexene, >99% purity, and dimethyl sulfoxide (DMSO), >99.9% purity, were purchased from AnalytiChem. Hexabromocyclododecane was kindly provided by Neue Materialien Bayreuth GmbH. An extruded polystyrene waste sample containing HBCD and hereafter denominated as “XPS waste” was obtained from a waste collector supplier. The XPS waste was a non-washed sample containing some visible dirt. The waste sample was used as received, only big visible metal and wood parts were removed manually and then the boards were shredded using an 8 mm sieve (Shini Plastic Technologies, SG-1635 N-CE) to obtain a smaller particle size in the range of 1.0 mm to 2.36 mm.

2.2. Solvent screening

Solvent selection is a key aspect for the success of liquid-liquid extraction. In this work, the goal is to keep polystyrene dissolved in the solvent (S1) whilst HBCD diffuses to the extractant solvent (S2), Fig. 1. A solvent combo has to be selected such that the three identified criteria are met, namely: (1) the solubility of polystyrene in the solvent (S1) has to be higher than in the extractant solvent (S2), (2) the solubility of HBCD in the solvent (S1) has to be lower than in extractant solvent (S2), and (3) the solvent (S1) and the extractant solvent (S2) need to be immiscible, Fig. 1.

To meet these criteria, a stepwise methodology was employed to obtain the solvent combination for the extraction of HBCD from polystyrene, Fig. 2. First, the solvents for polystyrene were selected based on the solvent screening performed in our previous work [14]. In this previous work we have applied the Hansen Solubility Parameters to a database of 455 solvents followed by experimental validation, which ensured a thorough screening, and several criteria were applied to reduce the solvents list, including the exclusion of halogenated solvents and solvents restricted under the REACH regulation (EC 1907/2006), such as benzene. Six of the top solvents for polystyrene identified in our previous work [14] were selected, Fig. 2. Top solvents in this context refer to solvents that allowed fast dissolution.

Next, COSMO-RS simulations were performed to select promising solvents for HBCD, from which solvents were identified as top solvents based on the (relative) solubilities obtained from COSMO-RS. Simultaneously, the (relative) solubilities for the same solvents were obtained for PS with COSMO-RS which allowed the assessment of the fulfilment of criteria 1 and 2.

COSMO-RS can predict both absolute and relative solubility. Absolute solubility provides a quantitative estimate of the solute's equilibrium concentration in a given solvent, whereas relative solubility compares the solvation affinity of a solute across different solvents by avoiding solid-state thermodynamic data (melting temperature and heat of fusion) and depends only on solute-liquid interactions via infinite dilution chemical potential, which is expressed through Eq.1:

$$x_i^{RS} = \exp\left(\frac{\mu_{i,\text{solvent}}^\infty}{RT}\right) \quad (1)$$

Where x_i^{RS} is the relative solubility, μ_i^∞ is the infinite dilution activity coefficient, R is the gas constant and T is temperature. Thus, solvents with lower (more favorable) infinite-dilution chemical potentials correspond to higher predicted solubility. For solvent screening purposes, relative solubility was used, as the primary objective was to compare the affinity of HBCD and PS across different solvents rather than determine their exact solubility values in a specific solvent. This resulted in a continuous work loop of checking which solvent combination fulfilled criteria 1 and 2. Finally, once potential solvent combos were identified, the last step was to check the miscibility of the

identified potential solvent combination using the (maximum) miscibility obtained from COSMO-RS and experimental validation. The experimental validation of solvents miscibility was performed by adding the two solvents at the same ratio in a vial at room temperature, followed by manual shake and visual determination of the presence or absence of two phases.

When needed, to support the solvent screening, the Hansen Solubility Theory was used. The Hansen Solubility Parameter (HSP) of the solvents and of polystyrene were taken from the literature [16] as well as of PS, the latter being $\delta_D = 21.3 \text{ MPa}^{1/2}$, $\delta_P = 5.8 \text{ MPa}^{1/2}$, $\delta_H = 4.3 \text{ MPa}^{1/2}$. The Relative Energy Difference (RED) value from the Hansen Solubility Theory was calculated following Kol et al. [14]. The HSP for HBCD are to the best of our knowledge not tabulated in literature, thus Quantitative Structure-Property Relationship (QSPR) based model in COSMO-quick was applied by using a non-optimized mol file directly acquired from ChemicalBook, which gave the following HSP parameters: $\delta_D = 16.401 \text{ MPa}^{1/2}$, $\delta_P = 7.146 \text{ MPa}^{1/2}$, $\delta_H = 7.646 \text{ MPa}^{1/2}$.

For the COSMO-RS simulations, HBCD was geometrically optimized in Gaussian16 using the B3LYP functional with Grimme's D3 dispersion correction and the 6-311++G(d,p) basis set. The molecular charge density required for COSMO-RS σ -profile generation was obtained from self-consistent reaction field (SCRF) calculations employing the COSMO solvation model. Thermal corrections were added to the COSMO file of HBCD, while COSMO files of polystyrene and all solvents were taken directly from the COSMOthermX TZVPD-FINE database. Using this data, the relative and absolute solubility of HBCD and polystyrene in the selected solvents was calculated at 25 °C.

2.2.1. Experimental solubility determination of PS and HBCD in the solvents

The solubility of polystyrene and HBCD in the selected solvents (cyclohexene and dimethyl sulfoxide), was experimentally tested. This is important since PS should be kept dissolved in the cyclohexene-rich phase whilst HBCD should diffuse to the dimethyl sulfoxide-rich phase. The solubility of PS in cyclohexene, which has been determined in our previous study [14], is defined as the entanglement concentration, since at higher concentration a gel is formed. The solubility determination of PS in dimethyl sulfoxide was done by preparing approximately 10 wt% solutions in a closed vial at 25 °C. Then, after stirring for about 24 h, the liquid phase was centrifuged at room temperature using a Hermle Z306 centrifuge at 4500 rpm for 20 min. The supernatant was analyzed with thermogravimetric analysis (TGA). The TGA measurements were carried out in a Netzsch TG 209 F3 Tarsus thermogravimeter under a constant flow of dry nitrogen at a rate of 60 mL·min⁻¹. The temperature profile started with a dynamic step from 35 to 250 °C at 15 °C·min⁻¹, followed by an isothermal at 250 °C for 15 min and a dynamic step up to 600 °C at 100 °C·min⁻¹, with a final dynamic step at 20 °C·min⁻¹ until 700 °C. The solubility of HBCD in cyclohexene has been reported in our previous study [6] and the same methodology was used to determine the solubility of HBCD in dimethyl sulfoxide. Duplicates were performed to ensure reproducibility.

2.2.2. Turbidity measurements

Turbidity measurements were performed to experimentally quantify solvent miscibility and to assess how variations in miscibility affect polymer precipitation. For the miscibility between solvents, dimethyl sulfoxide (or cyclohexene) was placed in a vial inside a water bath at 25 °C. Next, cyclohexene (or dimethyl sulfoxide) was added stepwise in fixed amounts in intervals of 30 min. Before each turbidity measurement, the solutions were placed in an ultrasonic bath for 1 min at 25 °C to ensure dispersion. The turbidity was monitored with a turbidimeter (model HI98703 by Hanna Instruments). Three measurements were performed and the average value is presented. The cloud-point was determined by analyzing the change in turbidity and complemented with visual determination when needed.

The same methodology was applied to analyze the phase separation

of dissolved polystyrene in the presence of the extractant. For this, solutions of 10 wt% polystyrene in cyclohexene were prepared and dimethyl sulfoxide was added stepwise. Duplicates were prepared.

2.3. Tie-line determination and partition coefficient

2.3.1. Experimental determination of tie-lines and partition coefficient

To determine the tie-lines, individual samples at various compositions of the three components (HBCD, cyclohexene and dimethyl sulfoxide) within the two-phase region were prepared by weight. The solutions were stirred for five hours at constant stirring in a controlled temperature bath at 25 °C and then were allowed to reach equilibrium by phase separation overnight at 25 °C. Two phases were formed, an upper phase which consisted of a cyclohexene-rich phase ($\rho_{\text{cyclohexene}} = 0.811 \text{ g}\cdot\text{cm}^{-3}$ at 25 °C) and a bottom phase which consisted of dimethyl sulfoxide-rich phase ($\rho_{\text{DMSO}} = 1.1 \text{ g}\cdot\text{cm}^{-3}$ at 25 °C). Next, aliquots from the upper and bottom phase were collected and analyzed with HPLC, based on a set of calibration curves for HBCD, cyclohexene and dimethyl sulfoxide (SI, Figs. S1-S3). For each sample, three measurements were performed.

HPLC analyses were performed on a Shimadzu LC-2050C series set to Shimadzu ELSD-LTIII detector. Chromatograms were recorded at 225 nm, and peaks were integrated using Labsolution software Agilent Pursuit C18 (150 × 2.0 mm 5 µm) and the Pursuit XR5 C18 MetaGuard (10 × 2.0 mm) guard column. Column thermostat was set to 40 °C and injection volume was set to 1 µL. The mobile phase consisted of 80%/20% acetonitrile/water and was used at a flow rate of 0.3 mL/min. A model sample was also prepared to quantify the error associated with the HBCD and dimethyl sulfoxide quantification. The result indicated that the standard deviation is 0.001 wt% for HBCD and 0.01 wt% for dimethyl sulfoxide (SI, Table S1).

The partition coefficient, K_p , is the ratio of the concentration of a substance (in this work, HBCD) in the extract phase to the raffinate phase at equilibrium, and was calculated for each tie-line using Eq. 2 [17,18]. The presence of an azeotrope between cyclohexene and dimethyl sulfoxide was analyzed using Aspen Plus.

$$K_p = \frac{w_{\text{HBCD in extract phase}}}{w_{\text{HBCD in raffinate phase}}} \quad (2)$$

where $w_{\text{HBCD in extract phase}}$ is the concentration of HBCD in the extract phase (dimethyl sulfoxide-rich phase in this case) and $w_{\text{HBCD in raffinate phase}}$ is the concentration of HBCD in the raffinate phase (cyclohexene-rich phase in this work).

The removal efficiency, RE, of HBCD for each tie-line was calculated using Eq. (3).

$$RE = \frac{w_{\text{HBCD in feed solution}} - w_{\text{HBCD in raffinate phase}}}{w_{\text{HBCD in feed solution}}} \bullet 100\% \quad (3)$$

where $w_{\text{HBCD in feed solution}}$ is the concentration of HBCD in the prepared (feed) solution and $w_{\text{HBCD in raffinate phase}}$ is the concentration of HBCD in the raffinate after extraction.

2.3.2. Determination of partition coefficient by COSMO simulations

Ternary liquid-liquid equilibrium (LLE) calculations were performed using the liquid extraction module of COSMOtherm. The initial system was defined as two liquid phases. In the first phase, the initial masses of dimethyl sulfoxide and HBCD were specified, while cyclohexene was placed in the second phase. The solvents, dimethyl sulfoxide and cyclohexene, were taken from the COSMOtherm TZVPD-FINE database. The HBCD COSMO file was generated following the computational protocol described in Section 2.2. Liquid extraction calculations were then performed to determine the equilibrium distribution of the components between the two phases, enabling prediction of the ternary LLE behavior of the HBCD–cyclohexene–dimethyl sulfoxide system. Subsequently the $K_{p,\text{COSMO}}$ values (reported in Section 3.2.) are calculated

based on estimated weight fraction of HBCD in extract and raffinate phase according to Eq. 2.

2.3.3. Simulations in ASPEN to calculate the number of stages

The extraction column was simulated by using the “Extract” module in Aspen Plus. The liquid–liquid equilibrium behavior of the HBCD–cyclohexene–dimethyl sulfoxide system, was modeled using the Liquid–Liquid Distribution Coefficient approach. The built-in model in Aspen Plus follows a general expression, Eq. (4).

$$\ln(K_p) = a + \frac{b}{T} + c \ln(T) + dT \quad (4)$$

Where a , b , c , and d are input parameters required from Aspen Plus, and T is the absolute temperature (K). In this study, the experiments were conducted at a single temperature; therefore, parameters b , c , and d were set to zero. A constant partition coefficient was assumed throughout the column and was set equal to the arithmetic mean of all experimentally determined partition coefficients (K_p) reported in Section 3.2. The solvent-to-feed ratio and number of theoretical stages were adjusted to meet the regulatory requirement of a maximum HBCD concentration of 100 ppm [15] in recycled polystyrene (PS). The HBCD concentration of 0.24 wt% in the feed was selected as the arithmetic mean of the expected HBCD content in polystyrene (0.8–4.0 wt%) [19]. Considering that the dissolution mixture contains 10 wt% polymer, this corresponds to an HBCD concentration of 0.08–0.40 wt% in solution, with 0.24 wt% representing the average composition of a heterogeneous waste stream. The HBCD concentration in the raffinate phase was targeted to be reduced to 10 ppm to ensure compliance with the specified legal limit.

2.4. Case-study: Extraction of HBCD from XPS construction waste

The extraction of HBCD from real polystyrene waste was performed. First, the XPS waste sample was analyzed with microwave-induced combustion (MIC) followed by ion chromatography to quantify the amount of Br present, and consequently of HBCD. For this, 20 mg of solid XPS waste was weighed and placed on the quartz holder layered with ashless filter paper inside a MIC vessel (Multiwave 5000, Anton Paar). Subsequently, 8 mL of a solution was added, which consisted of distilled water with 8 mM Na_2CO_3 and 1 mM NaHCO_3 . Next, the samples were loaded onto a carousel and a pressure of 20 bar O_2 was applied. The combustion program was as follows: the power is set to 900 W for 10 min, next, the temperature and the pressure increase during those 10 min to approximately 230 °C and 65 bar, and the final step is an isothermal for 10 min. Upon completion of the combustion, the content was transferred to a graduated flask with a final of 25 mL, to which distilled water was added to dilute the content. Next, the solution was filtered and analyzed with ion chromatography using an ECO IC 1 system (Metrohm) coupled with a 863 Compact Autosampler 1, equipped with a peristaltic pump. Separation was achieved using a Metrosep A Supp 19–150/4.0 analytical column (length 150.0 mm, inner diameter 4.0 mm, particle size 4.6 µm), preceded by a Metrosep A Supp 19 Guard/4.0 pre-column (length 5.0 mm, inner diameter 4.0 mm). The flow rate of the eluent was maintained at $0.7 \text{ mL}\cdot\text{min}^{-1}$, with a system pressure of approximately 13.4 MPa. The initial calibration curve was constructed using six standard solutions with bromine concentrations of 0.05, 0.1, 0.25, 1, 3 and 5 mg/L.

For the extraction experiments, a 10 wt% XPS waste solution in cyclohexene was prepared at room temperature. After the polymer was dissolved, the solution was added to a separation funnel, and dimethyl sulfoxide was added in a 1:1 volume ratio to the solvent at room temperature. The solution was vigorously shaken and left still for 24 h. Next, the two phases were separated. The experiment was done in duplicate. The upper phase was analyzed using microwave-assisted combustion followed by ion chromatography. The procedure was identical as for the solid XPS waste sample, with the only difference that cyclohexene was

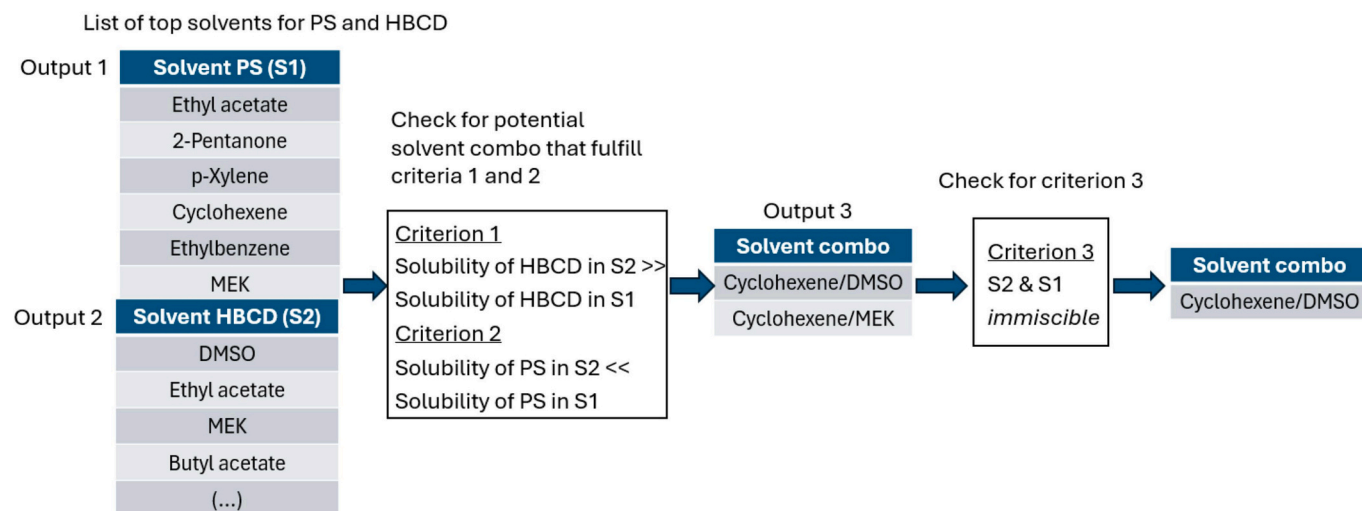


Fig. 3. Results from solvent screening using the stepwise methodology, with S1 being the solvent and S2 being the extractant solvent.

Table 1

COSMO-RS (absolute) solubilities, (absolute) miscibility and experimental solubilities and miscibility with selected solvents at 25 °C.

Solvent	Cyclohexene (CH)
Extractant	Dimethyl sulfoxide (DMSO)
RED value for PS in solvents	RED _{PS-cyclohexene} = 0.75 RED _{PS-DMSO} = 1.06
M_{COSMO-RS} of solvents	M _{cyclohexene/DMSO} = 7.3 wt% M _{DMSO/cyclohexene} = 3.4 wt%
S_{COSMO-RS} of PS in solvents	S _{PS/cyclohexene} = NA* S _{PS/DMSO} = NA*
S_{COSMO-RS} of HBCD in solvents	S _{HBCD/cyclohexene} = 0.74 wt% [6] S _{HBCD/DMSO} = 53.6 wt%
M_{exp} of solvents	M _{cyclohexene/DMSO} = 10.0 wt% M _{DMSO/cyclohexene} = 1.07 wt%
S_{exp} of PS in solvents	c _{e,PS/cyclohexene} = 13.8 wt% [14] S _{PS/DMSO} = 0 wt%
S_{exp} of HBCD in solvents	S _{HBCD/cyclohexene} = 1.10 ± 0.10 wt% [6] S _{HBCD/DMSO} = 13.9 ± 0.3 wt%

RED value – relative energy difference value from Hansen Solubility Theory, c_e – entanglement concentration, $S_{\text{COSMO-RS}}$ – absolute solubility obtained by COSMO-RS, $M_{\text{COSMO-RS}}$ – absolute miscibility obtained by COSMO-RS, S_{exp} – experimental solubility, M_{exp} – experimental miscibility. *COSMO-RS retrieves NA, meaning that solubility is too high.

left to evaporate prior to the addition of the sample to the MIC. The removal efficiency, RE, of HBCD was calculated using Eq. 3 presented in Section 2.3.1.

3. Results

3.1. Solvent selection for liquid – liquid extraction

Based on the methodology presented for solvent selection in Section 2.2, six top solvents were identified for polystyrene, being cyclohexene, 2-pentanone, ethylbenzene, methyl ethyl ketone (MEK), ethyl acetate and xylene, Fig. 3. For HBCD top solvents were selected from the COSMO-RS simulations, with the top four being dimethyl sulfoxide, ethyl acetate, methyl ethyl ketone and butyl acetate, Fig. 3, ranked from highest (relative) to lowest (relative) solubility obtained from COSMO-RS, (SI – Tables S2-S3). After comparing the (relative) solubilities of PS and HBCD in S1 and S2, two solvent combos were identified, namely cyclohexene and DMSO and cyclohexene and MEK. For both, PS has a higher (relative) solubility in cyclohexene (5.127) than in DMSO (3.375) or MEK (4.984) and HBCD has a higher (relative) solubility in DMSO (6.523) and MEK (6.266) compared to cyclohexene (3.991), (SI –

Tables S2-S3). Following this, the (absolute) miscibility simulation from COSMO-RS indicates that cyclohexene and dimethyl sulfoxide have low miscibility, Table 1. For cyclohexene and MEK no COSMO-RS data was available, but the experimental results indicated that these two solvents were miscible. Thus, from this solvent screening, cyclohexene and dimethyl sulfoxide came out as the solvent combination that fulfilled all the identified criteria, with DMSO being the extractant solvent, Fig. 3.

Cyclohexene, identified as a top solvent in our previous work for polystyrene [14], dissolves the polymer at room temperature, with an entanglement concentration at 13.8 wt%. In contrast, the RED value for polystyrene-DMSO is of 1.06, which indicated that DMSO was outside the radius of interaction of polystyrene and thus does not likely dissolve PS, Table 1. The experimental results confirm that PS does not dissolve in DMSO (SI – Figs. S4 – S5).

The experimental results confirmed that the solubility of HBCD in DMSO is higher than in cyclohexene, namely of 13.9 ± 0.3 wt% (SI – Figs. S6 – S7) and 1.10 ± 0.10 wt% [6], respectively, as seen in Table 1. The solubility of HBCD in cyclohexene is quite low, especially compared to its solubility in other solvents such as butyl acetate (7.92 ± 0.42 wt%)

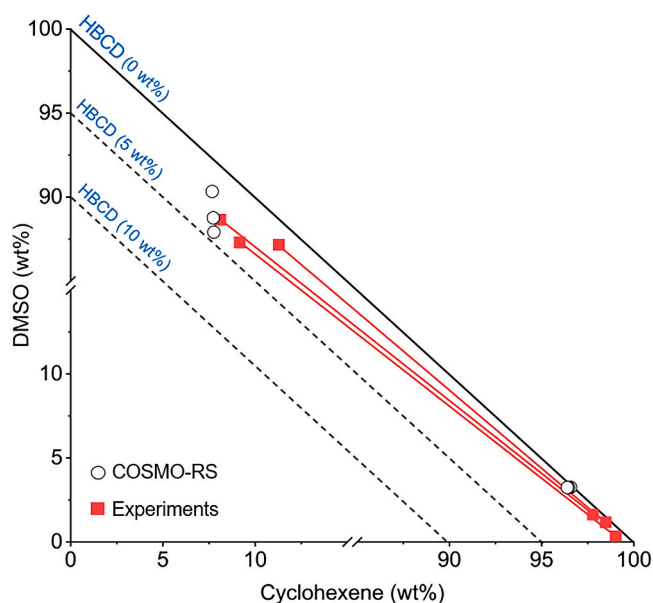


Fig. 4. Rectangular ternary phase diagram of HBCD-cyclohexene-dimethyl sulfoxide system at T = 25 °C.

Table 2

Weight concentration of the experimental and COSMO-RS tie-lines of the ternary system HBCD–cyclohexene–dimethyl sulfoxide at $T = 25\text{ }^{\circ}\text{C}$, along with the experimental K_p ($K_{p,exp}$), K_p obtained by COSMO-RS ($K_{p,COSMO-RS}$) and experimental HBCD removal efficiency for each tie-line mixture.

Mixture	Cyclohexene (CH)-rich phase			Dimethyl sulfoxide (DMSO)-rich phase				
	HBCD (wt%)	CH (wt%)	DMSO (wt%)	Experimental			$K_{p,exp}$ (/)	RE (%)
				HBCD (wt%)	CH (wt%)	DMSO (wt%)		
1	0.361 ± 0.004	98.471 ± 0.007	1.168 ± 0.003	1.562 ± 0.006	11.28 ± 0.01	87.16 ± 0.01	4.3	71.9 ± 0.3
2	0.585 ± 0.006	97.79 ± 0.04	1.63 ± 0.03	3.24 ± 0.06	8.11 ± 0.03	88.65 ± 0.03	5.5	73.2 ± 0.3
3	0.67 ± 0.01	99.007 ± 0.008	0.318 ± 0.003	3.54 ± 0.02	9.16 ± 0.02	87.29 ± 0.02	5.3	74.6 ± 0.4

Mixture	Cyclohexene (CH)-rich phase			Dimethyl sulfoxide (DMSO)-rich phase			
	HBCD (wt%)	CH (wt%)	DMSO (wt%)	COSMO-RS			$K_{p,COSMO-RS}$ (/)
				HBCD (wt%)	CH (wt%)	DMSO (wt%)	
1	0.1764	96.5771	3.2465	2.0039	7.6708	90.3252	11.36
2	0.3047	96.4552	3.2402	3.5130	7.7251	88.7619	11.53
3	0.3651	96.3973	3.2376	4.3286	7.7576	87.9138	11.85

[6]. However, flame retardants are expected to be in the range of 0.8–4 wt% in the polymer [19], thus, for a 10 wt% polymer solution, this corresponds to a range of 0.08–0.4 wt% of HBCD in solution, which is well below the solubility limit of HBCD in cyclohexene. Thus, this indicates that HBCD is dissolved when having 10 wt% polymeric solution and cannot be separated by solid-liquid separation processes such as filtration or centrifugation, as observed for other flame retardants [6].

Finally, experimental runs (SI – Figs. S8–S9) confirmed that the two solvents have low miscibility, Table 1, with cyclohexene dissolving in higher amount in dimethyl sulfoxide (10.0 wt%) than dimethyl sulfoxide in cyclohexene (1.07 wt%, respectively), Table 1. The results also show that the addition up to 1.07 wt% dimethyl sulfoxide to the solution of dissolved polystyrene in cyclohexene does not lead to polymer precipitation (SI – Fig. S10).

In sum, the simulations and experimental results confirm that the combo cyclohexene-dimethyl sulfoxide fulfills the three required criteria for the extraction, namely (i) HBCD has a higher solubility in dimethyl sulfoxide than in cyclohexene, (ii) PS has a higher solubility in cyclohexene than in dimethyl sulfoxide and (iii) dimethyl sulfoxide and cyclohexene have low miscibility.

3.2. Liquid-liquid equilibrium data

The experimentally determined ternary phase diagram, including its tie-lines, are displayed in Fig. 4. The weight concentrations of the prepared solutions of HBCD, cyclohexene and dimethyl sulfoxide, together with the partition coefficient are presented in Table 2. The results show that the biphasic region is wide, indicating that the two solvents have limited miscibility, which results in an efficient separation into two distinct phases. Due to difficulty obtaining HBCD as a molecule for the experiments, the zones with higher HBCD content and the binodal curve were not determined experimentally in the phase diagram. These zones would also be less relevant for this work, as expected HBCD concentration in dissolution recycling is below 1 wt% in 10 wt% PS solutions. The tie-lines obtained with COSMO-RS are depicted in Fig. 4, indicating a low discrepancy of less than 2 wt% between experimental data and COSMO-RS simulations. The experimentally obtained K_p values are in the range between 3.5 and 5.5, indicating that the extraction of HBCD into the DMSO phase is favorable, Table 2. The K_p values obtained by COSMO-RS are higher and possibly overestimated, Table 2, which may be related to the low solubility of HBCD in cyclohexene. Small inaccuracies in the predicted solubility can result in large deviations in the estimation of the K_p values with COSMO-RS [20,21], as the K_p calculation is also sensitive to variation in the denominator due to the relatively

large numerator/denominator ratio. Dimethyl sulfoxide and cyclohexene also do not present an azeotrope, which can facilitate the recovery of the solvents downstream (SI - Fig. S11).

3.3. Extraction column simulations

Finally, to forecast the performance of HBCD removal when moving from laboratory experiments to an extraction column design proposal, a simulation was developed in Aspen. The feed stream was assumed to come from the polymer dissolution step and consisted of 10 wt% polystyrene (PS), 0.24 wt% HBCD, and 89.76 wt% cyclohexene (see details in Section 2.3.3. and Section S6 of Supporting Information for the assumed concentrations). To assess the extraction process, the number of theoretical stages in the extraction column was varied from 1 to 10, while the solvent-to-feed ratio was varied from 0 to 1.5 (g g^{-1}), to evaluate their effects on the HBCD concentration in the raffinate stream (Fig. 5a). Subsequently, for each column configuration with 2–10 theoretical stages, the DMSO flow rate was adjusted to achieve the target HBCD concentration of 10 ppm in the raffinate stream (Fig. 5b).

Fig. 5a shows that a minimum of three theoretical stages is required. Under these conditions, the solvent-to-feed ratio is approximately 0.9 g g^{-1} . Increasing the number of stages reduces the solvent demand. For example, at ten stages, the required solvent decreases to around 0.24 g g^{-1} . The relationship between the number of stages and the minimum solvent requirement is presented in Fig. 5b, showing that beyond approximately 4–5 stages only limited reductions in solvent requirement are achieved. Yet, the simulation is based on a constant average partition coefficient obtained for the ternary system HBCD–cyclohexene–dimethyl sulfoxide, while the partition coefficient is expected to vary with the local phase compositions throughout the extraction column and in the presence of polymer (and potentially other additives in the case of a real waste sample). A more rigorous liquid–liquid equilibrium model, such as NRTL, would be required to capture this behavior, but the quantity of equilibrium data needed for model regression are currently unavailable. Therefore, the predicted number of stages and solvent requirements should be interpreted as preliminary design estimates. Based on these results, a simple solvent management process was proposed and incorporated into the Aspen Plus simulation (Fig. 5c). The process is operated at a feed polystyrene flow rate of 100.00 kg h^{-1} . After extraction, the DMSO-rich extract stream is first sent to a distillation column, where the dissolved cyclohexene is recovered as the overhead product with a recovery of 3.0% and a condenser and reboiler duty of 6.43 kW and 29.11 kW, respectively. The distillation column was designed to limit the dimethyl sulfoxide content in the overhead stream to no more than 2 wt

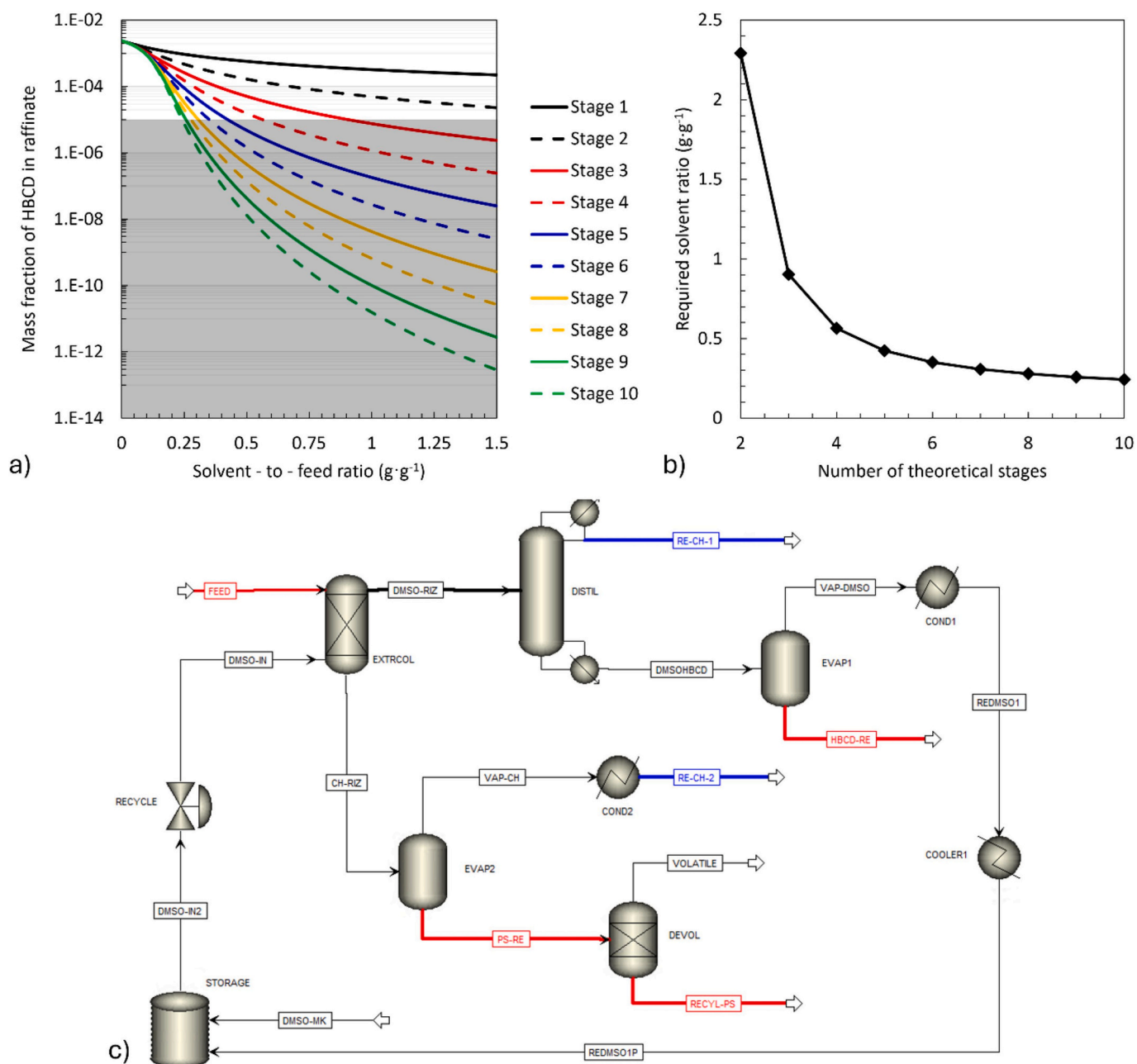


Fig. 5. (a) Variation of HBCD concentration in raffinate with theoretical extraction stage and solvent-to-feed ratio. The gray shaded region represents the attainable raffinate concentration below 10 ppm. (b) Required solvent quantity to achieve the target HBCD concentration. (c) Process flow diagram obtained from Aspen Plus including solvent management.

%, allowing high-purity cyclohexene (> 98 wt%) to be recycled to the dissolution step. The bottom stream, mainly containing DMSO and HBCD, is subsequently fed to an evaporator, where DMSO is vaporized and recovered at 99.99 wt% purity for recycling to the extraction column, while HBCD is recovered in the bottom product, with a heat duty of 40.64 kW. The raffinate stream, consisting primarily of cyclohexene and polystyrene, is also sent to an evaporator to recover the recycled polymer containing less than 100 ppm HBCD. The evaporated recovered cyclohexene (recovery of 96.9 wt% to the feed cyclohexene) contains less than 2 wt% dimethyl sulfoxide, allowing it to be recycled directly to the dissolution step. The DMSO flow rate in the system is of $250.00 \text{ kg} \cdot \text{h}^{-1}$ with a make-up solvent flow rate of $6.66 \text{ kg} \cdot \text{h}^{-1}$. The cyclohexene flow rate in the feed is of $897.60 \text{ kg} \cdot \text{h}^{-1}$ and cyclohexene is fully recovered in the system. This process was simulated based on the case with a 10 theoretical stage extraction column. The total heating demand for the

solvent recovery is the sum of the evaporator and reboiler heat duties, giving a total of 192.72 kW. The cooling demand is the sum of the absolute values of condenser and cooler heat duties, giving a total of 163.09 kW. The total utility demand for the solvent recovery is then of 355.81 kW (see SI – Section S6). Considering the feed polystyrene flow rate in the process, this corresponds to a total utility demand of 3558.1 kWh per ton of feed polystyrene for the solvent recovery process. To further reduce the energy demand, heat integration strategies should be implemented, which were not included in this assessment. The detailed simulation setup, together with the corresponding heat and mass balances, is provided in the Supporting Information (Section S6).

3.4. Case-study with waste XPS

The proposed liquid-liquid extraction was performed on a real waste

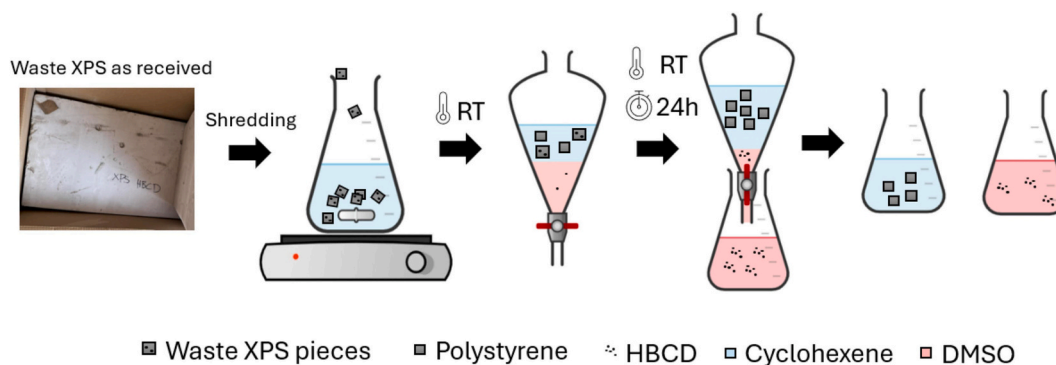


Fig. 6. Scheme of liquid-liquid extraction experiments performed on a real waste polystyrene insulation board sample.

sample, namely a polystyrene insulation board containing HBCD, Fig. 6. The results indicate that the waste sample contains 15,908 mg/kg of HBCD, which is within the expected flame retardant concentration range of 0.8–4 wt% in the polymer [19]. Additionally, a removal efficiency of $93.7 \pm 0.7\%$ was obtained with one extraction step with a DMSO – to – feed ratio ($\text{g} \cdot \text{g}^{-1}$) of 1.2 (the result details can be found in Supporting Information, Section S7). This value is higher than the removal efficiencies expected from the LLE experiments in Section 3.2., with the highest obtained removal efficiency being of $74.6 \pm 0.4\%$ for the third mixture in Table 2. When compared to literature, this is a higher removal efficiency for HBCD than the removal efficiency that Mosallaei et al. [13] obtained with supercritical CO_2 /co-solvent mixture extraction and similar to the extraction efficiencies obtained by Vilaplana et al. [9] using pressurized liquid extraction and microwave-assisted extraction with isopropanol/n-hexane solvent mixture at temperatures higher than 100°C . Yet, the sustainability and economic feasibility of all these techniques cannot be compared without the performance of a Life Cycle Assessment and Techno-Economic Analysis, which is out of scope from this work. Furthermore, the results obtained in this work indicate that the presence of the polymer, and potentially other additives, facilitates the extraction of HBCD. This may be related to the fact that the partitioning of a solute between two phases is governed by differences in chemical potential and mass transfer of the solute occurs between phases until thermodynamic equilibrium is reached, at which point the chemical potential of the solute in both phases is the same [17,22]. The addition of another component into the system may alter the activity coefficient and consequently the partition coefficient [23]. Thus, the results indicate the addition of PS or the presence of other additives may be leading to a change of the activity coefficient and consequently of the partition coefficient, resulting in enhanced diffusion and partitioning of HBCD into the DMSO phase. These results indicate that the obtained ternary equilibrium data does not fully capture the behavior in real waste dissolution processes, posing a limitation to quantitatively predict HBCD removal from real XPS waste and to perform process simulations. Hence, as future work, it would be relevant to obtain liquid-liquid equilibrium data for a quaternary system and subsequently partition coefficient containing four components and ideally, other relevant additives such as stabilizers, colorants, and residual processing aids present in the waste stream [24]. Finally, it should be noted that both dimethyl sulfoxide and cyclohexene are registered under the REACH regulation [25,26]. Dimethyl sulfoxide is not considered a hazardous substance according to Regulation (EC) No. 1272/2008 and does not have restrictions in use [26–28], yet, cyclohexene is flammable, toxic and hazardous to the aquatic environment [25]. Dimethyl sulfoxide has a flash point of 87°C and cyclohexene of -20°C , which means that the full process must be operated in compliance with ATEX requirements, which increases capital and operating costs. From an environmental perspective, cyclohexene should be handled in a closed system to prevent emissions and release to the environment. To minimize solvent waste generation, both solvents should be efficiently recovered and

reused. As future work, the polymer recovery step should also be tested, as well as comprehensive Life Cycle Assessment and Techno-Economic Analysis should be performed to quantify the overall environmental impact and economic feasibility.

4. Conclusions

In this work we have demonstrated the successful removal of HBCD using liquid-liquid extraction at 25°C with the solvent combination of cyclohexene and dimethyl sulfoxide. The liquid-liquid equilibrium data for the ternary system of HBCD-cyclohexene-dimethyl sulfoxide was constructed showing a wide biphasic region and the obtained partition coefficients confirm that DMSO has a strong affinity for HBCD, which confirms its suitability as extractant solvent. Finally, the success of HBCD removal was demonstrated with a real polystyrene insulation board waste sample. The result shows that the presence of polymer facilitates the extraction, with ASPEN simulations indicating that a minimum of three stages and a solvent-to-feed ratio of 0.9 is needed to achieve the legal limit of 100 ppm when considering a feed solution containing 10 wt% dissolved polymer. Therefore, this paper is the first evidence that liquid-liquid extraction might be a feasible additive removal technique in dissolution recycling, within the specific context of the extraction of the legacy chemical HBCD from polystyrene insulation waste.

CRediT authorship contribution statement

Rita Kol: Conceptualization, Methodology, Visualization, Validation, Formal analysis, Investigation, Project administration, Funding acquisition, Writing – original draft. **Amin Shariatmadar Tehrani:** Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing – original draft. **Thien Nguyen Luu Minh:** Software, Formal analysis, Writing – original draft. **Elina Selmurzaeva:** Investigation. **Norbert Niessner:** Writing – review & editing. **Steven De Meester:** Conceptualization, Visualization, Project administration, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Steven De Meester reports financial support was provided by Vlaams Agentschap Innoveren & Ondernemen and by Catalisti. Steven De Meester, Rita Kol and Norbert Niessner have patent pending to INEOS Styrolution Group GmbH. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.seppur.2026.139526>.

Data availability

Data will be made available on request.

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